RATTY2TAILS · ESSAY SERIES

On Education, Attention & the Machines We Hand Our Children

The Analog Rebellion

Why putting AI in the classroom may be a catastrophe — and why the world's schools are beginning to swim the other way.

An essay adapted from the address "Why AI in the Classroom is a Catastrophe", delivered by **Sophie Winkleman** at the Alliance for Responsible Citizenship (ARC) conference, 2026. Edited, annotated and set for print by Ratty2Tails.

Ratty2Tails

EDITOR & ANNOTATOR

Source: Sophie Winkleman, ARC 2026, London ExCeL.
Essay edition · British English throughout.

TL;DR

The short version

- A veteran head teacher found her pupils had, quite literally, **forgotten how to play** — their imaginations dulled by a childhood of constant, low-level screen stimulation. Boredom, Winkleman argues, is a free and vital catalyst for the imagination.
- Social media, smartphones and "EdTech" are **fins and teeth of the same shark**: the streaks, notifications and variable rewards flagged as harmful on social apps are the very same mechanics now baked into maths, language and even reading apps. They boost engagement, which is what the business model needs — not learning.
- We are **physical beings who learn physically.** Material read from a page implants more deeply than the same words on a screen; handwriting lights the brain up while typing leaves it comparatively dark. Cited research puts paper-based pupils roughly six months ahead of screen-based peers.
- The world is already turning: **Sweden, Denmark, Madrid, Glasgow, Los Angeles** and others are pulling back devices and reinvesting in textbooks — while many tech leaders quietly send their own children to low-tech schools.
- AI is the coming **"civilisational earthquake."** For an adult who has already learned to think, it is a powerful tool; for a child who has not, it skips the hard, formative "crawling phase" of cognitive development — trading immediate relief for long-term dependency.
- The prescription: **tech ed, not EdTech.** Teach how technology works in dedicated lessons; return to books, paper, pens and human teachers for everything else.

WHAT THIS IS

A faithful essay adaptation of Winkleman's ARC 2026 talk, restructured for reading, with the empirical claims flagged and, where possible, checked against public sources.

WHAT IT IS NOT

A blanket verdict that all classroom technology is worthless. Assistive tools for genuine disability are exempted; the target is default, universal screen-first schooling.

EDITORIAL STANCE

Ratty2Tails presents the argument on its own terms and hedges figures that could not be independently verified. Sources are listed at the end.

§ 1 — Introduction

The children who forgot how to play

A few months before she took the ARC stage, Sophie Winkleman was introduced to a head teacher — call her Justine — who runs a primary school in Hertfordshire. One sunny afternoon, on impulse, Justine cancelled lessons, opened the back doors and told the children to go outside and play. They looked at her, confused, and shuffled onto the grass. They stood there awkwardly for a few minutes, and then asked to come back in.

They could not, it turned out, generate their own fun. So thoroughly had entertainment been delivered to them, through glowing rectangles, that the machinery of unprompted play had seized up. These children, Winkleman is careful to note, were not the casualties of some horror encountered online. They were not traumatised. They were dazed, glazed and passive — young minds deactivated by a steady drip of low-level stimulation.

Justine's response was quietly radical. She scrapped screen-based story time and encouraged parents to let their children read at home — or simply be bored. Boredom, after all, costs nothing, and it is one of the most reliable catalysts the imagination has. A few weeks later, she reported that the same children now could not be stopped from tearing outside at break, their sociability and invention blossoming.

From that small parable Winkleman builds a larger, uncomfortable argument: that the very machines eroding children's attention and imagination at home have, at school, been installed as the principal conduits of their education. This essay follows that argument through its three movements — the case against **EdTech**, the deeper case for **embodied, analog learning**, and finally the gathering storm of **AI in the classroom** — and weighs the evidence offered for each.

> "Boredom costs nothing, and is a crucial catalyst for the imagination."
>
> — SOPHIE WINKLEMAN, ARC 2026

§ 2 — The sales pitch

EdTech: engagement dressed as education

Internet-enabled devices are handed out in schools across the world under the banner of a "future-proof" education. In Britain, Winkleman claims, around a million children now spend nearly every lesson on a screen. Many parents are not buying the pitch. In fast-growing online communities they ask the same questions over and over: Where are the books? Why is my child accessing obscene content on a school-issued iPad? Why is he online for homework every night, getting headaches, going short-sighted, sleeping later, drifting off task?

Some have moved from grievance to organised action. The US alliance **Schools Beyond Screens**, Winkleman recounts, persuaded the Los Angeles school district to approve one of the most sweeping screen-time pullbacks of any large public system in America. Its deputy director's summary is blunt: cognitive skills, attention spans and executive function are all declining; human brains do not learn well from screens; the research says one thing and the schools do another.

The heart of the EdTech critique is a claim about design. Social media, smartphones and educational apps, Winkleman argues, are not separate phenomena but "fins and teeth of the same shark." The features deemed harmful enough to restrict on social platforms — the streaks, the daily rewards, the endless notifications — reappear, almost unchanged, inside maths, language and reading apps. These are slot-machine tactics: the same unpredictable reward that keeps an adult scrolling, now wired into the schoolwork itself.

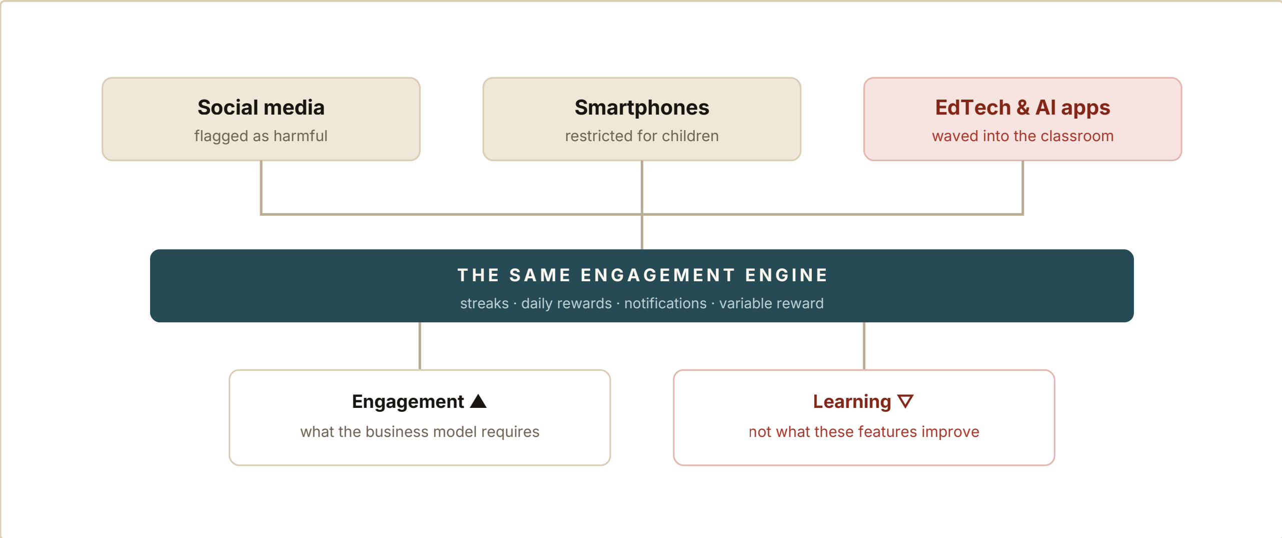

Figure 1. Fins of the same shark. Winkleman's core structural claim: the persuasive-design mechanics restricted on consumer apps are the same ones embedded in educational software. As the quoted critic Will Ellis puts it, such features exist "because they improve engagement, and engagement is what the business model requires" — not because they improve learning.

A reading app, she notes drily, may be the most oxymoronic product of all: a machine that gamifies the one activity whose whole value lies in sustained, undistracted attention.

§ 3 — The body knows

Why the page beats the screen

The distracting features, Winkleman argues, are not even the main problem. The deeper claim is about **embodiment**: that we are physical beings, and that we must therefore learn physically. Material read from a page in a book, on this account, implants in the brain more profoundly than the identical words read from a screen. The feel of the paper, the design of the cover, the typeface, the very smell of the pages, the spatial memory of where on the page a phrase sat — all of it contributes to a holistic, embodied act of learning.

Reading on a screen, by contrast, is both immaterial and aspatial. What you are reading now occupies the same physical place as what you read ten minutes ago, yesterday, last week. It is flat. It is, in her word, unimpactful — and so the information is less likely to stick. She cites the neuroscientist Jared Cooney Horvath on the brain's need for a period of "digestion" after learning: calm, book-based study grants this downtime as a matter of course, whereas screens, stimulating in and of themselves, never let the system rest.

Handwriting versus typing

The most striking evidence concerns handwriting. Winkleman draws on the work of the neuroscientist **Audrey van der Meer**, who has spent two decades charting the difference between writing by hand and typing. In high-density EEG studies, van der Meer and her colleague Ruud van der Weel found that writing words by hand produces widespread, elaborate connectivity across the brain, while typing the same words produces comparatively little.[2] Winkleman's image for it is vivid: the brain lit up "like a Christmas tree" during handwriting, and "dark as a cellar" during typing.

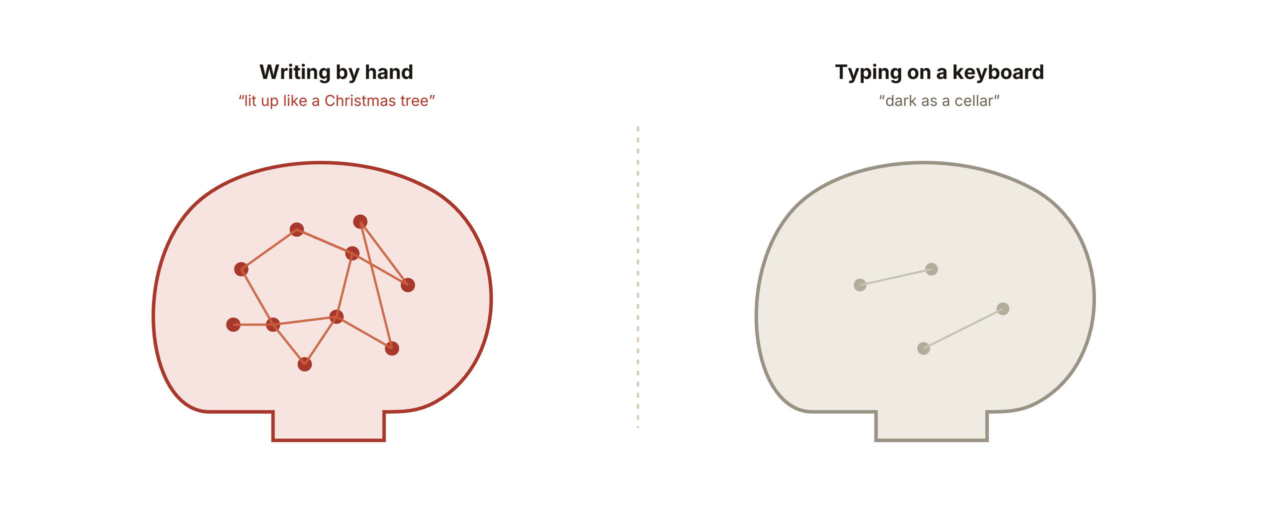

Figure 2. Handwriting versus typing. A schematic of the contrast Winkleman draws from van der Meer's EEG research: broad cross-brain connectivity when forming letters by hand, minimal equivalent activity when typing.[2] Handwriting is also slower — forcing a pupil to organise and prioritise what they hear — and it is personal, connecting the writer to the subject in a way a sterile typeface cannot. **(Illustrative schematic, not scan data.)**

Handwriting's very slowness is part of the point. Because the hand cannot keep pace with speech, a note-taking pupil must continuously decide what matters — a filtering discipline that transcriptive touch-typing renders unnecessary. And a teacher's handwritten comment on a returned piece of work, Winkleman adds, carries a human charge that robotically generated feedback cannot: personal, not processed, and essential to sustained motivation.

> "A child's handwriting connects him to the subject he's writing about, where a sterile typeface neutralises that relationship before it's even begun."
>
> — SOPHIE WINKLEMAN, ARC 2026

§ 4 — The scoreboard

The evidence on paper

The mathematics classroom fares no better under screens, in Winkleman's telling. She cites teachers and researchers who argue that working problems out on paper deepens mathematical understanding and confidence, and points to a controlled experiment — recounted in her address — in which some 3,000 pupils sat PISA-style tests in maths, science and reading. Half worked entirely on paper; half worked entirely on a computer. The paper group, she reports, scored twenty points higher: roughly the equivalent of six additional months of schooling.

EDITOR'S NOTE ON VERIFICATION

Ratty2Tails could not independently confirm the specific 3,000-pupil experiment or every name attached to it in the transcript, some of which appear garbled in transcription. The direction of the finding, however, is consistent with a substantial body of research on screen-versus-paper comprehension and with the OECD's own analyses linking heavy in-class device use to weaker outcomes. Treat the exact figures as claims made in the talk rather than as settled fact.

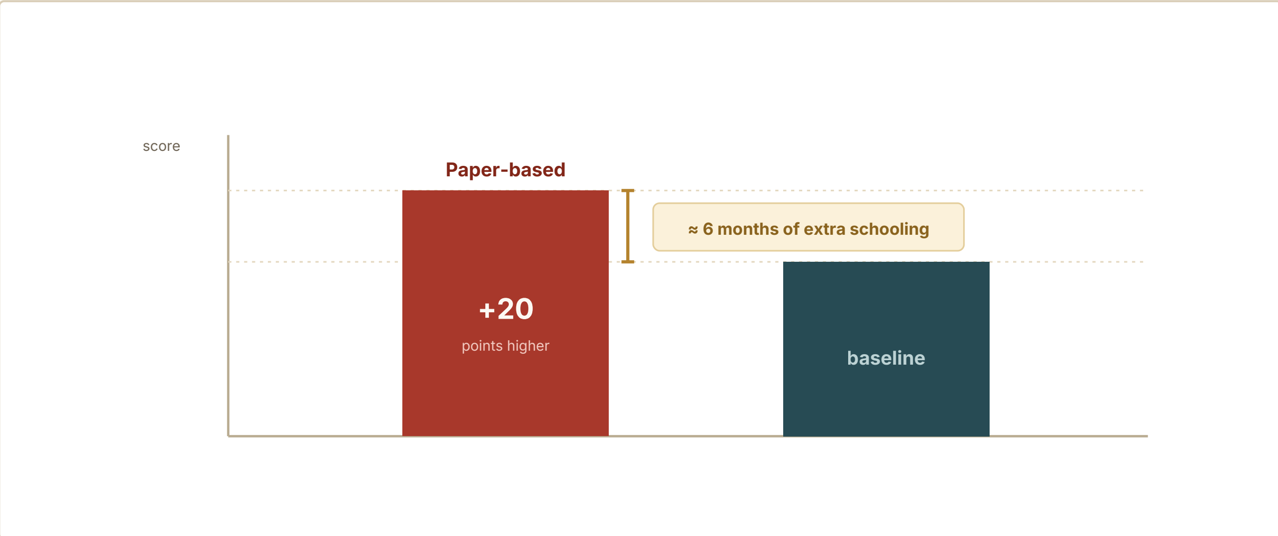

Figure 3. The paper premium, as cited in the talk. In the experiment Winkleman describes, pupils working on paper outscored those working on screen by about twenty points — a gap she equates to roughly six months of additional schooling. **Figures reproduce the claim as stated; see editor's note above.**

A world beginning to turn

Crucially, this is no longer a fringe worry. Winkleman lists a growing roster of places quietly reversing course. In **Madrid**, under-twelves no longer learn on devices. In **Glasgow**, the director of education binned a large maths app, reasoning that the work could be done on paper just as well while keeping the teacher close to the child. **Sweden and Denmark**, she notes, are spending heavily to reintroduce textbooks — a claim that checks out: Sweden alone committed on the order of €100 million to bring printed textbooks back after its PISA scores slid.[3] Equivalent pushbacks, she adds, are under way in Utah, Delaware and New York.

~1M	+20	€100M+	80%
UK children spending nearly every lesson on a screen (per the talk)	points for paper-based pupils ≈ 6 months of schooling	Sweden's reinvestment in printed textbooks[3]	of US middle schools issuing Chromebooks with built-in AI (per the talk)

These countries and cities, she suggests, are simply the first fish to break away from a misdirected shoal. The data of the past fifteen years, on her reading, shows scores beginning to fall around the world just as devices entered the classroom — a period that also, inconveniently for tidy causal stories, coincides with the explosion of social media into children's lives.

§ 5 — Nuance and irony

Two honest complications

The special-needs exception

Winkleman is careful not to overclaim. Some technology, she readily grants, is genuinely helpful: a speech-generating device for a non-verbal child, an eye-tracker for a physically impaired child, an audiobook for a blind child. But the small number of pupils truly served by assistive technology, she argues, should not dictate how the majority learn. For milder learning needs the picture is murkier still: the speech-and-language therapist she cites finds that classroom tech can disempower such children, making them dependent on a crutch precisely where they most need to exercise the skills they find hard. The push to make everything easier is seductive — and can quietly make things harder.

The tell-tale rebellion of the elite

Then comes the argument's sharpest irony. Many technology leaders, Winkleman observes, send their own children to conspicuously low-tech schools — aware, perhaps, that the most valuable human capital of the coming era will be the capacity to think deeply, knowledgeably and originally: to be the best of what a human can be, rather than a pale imitation of an AI. She recalls predicting, years ago and on the strength of a UNESCO report, that elite schools would turn their backs on screens and return to books while poorer children were left with the "digital slop."

> EdTech was meant to bridge the attainment gap. It seems, instead, to be widening it.
>
> — THE ESSAY'S CENTRAL IRONY

That prediction, she argues, is coming true: schools such as Winchester College adopt stringent paper-first policies, while some of the poorest children are handed chatbot tutors and a bouncing avatar in place of a human mentor. A machine, she insists, cannot be a mentor. It will not wonder how you are when you switch it off. Children facing difficulties at home deserve more human contact in the classroom, not less. Yes, human teachers and physical textbooks are expensive — but so is EdTech, a multi-billion-pound industry. The question is only where the money should go.

§ 6 — The earthquake

From EdTech tremor to AI earthquake

Everything so far, Winkleman says, describes only the tremor. The earthquake is AI. In its best form, she concedes, AI is a supremely capable tool — conjuring what you need from scratch, or sifting gigabytes into a cogent order. For a time-pressed adult who has already learned, the hard way, how to read, think, analyse and organise their own thoughts, that can be genuinely useful.

But a child has not yet made that long, fibrous, ultimately joyful journey. Hand them a tool that does the thinking, and they skip a crucial stretch of cognitive development. The behavioural economist Richard Thaler observed that humans default to the low-effort option; teenagers, testing every boundary, doubly so. However many earnest lectures they receive on using AI "responsibly" — as a sparring partner, not a second brain — most will take the shortest route to getting the essay done. That is not a moral failing. It is, Winkleman argues, a predictable biological response we have designed the system to invite.

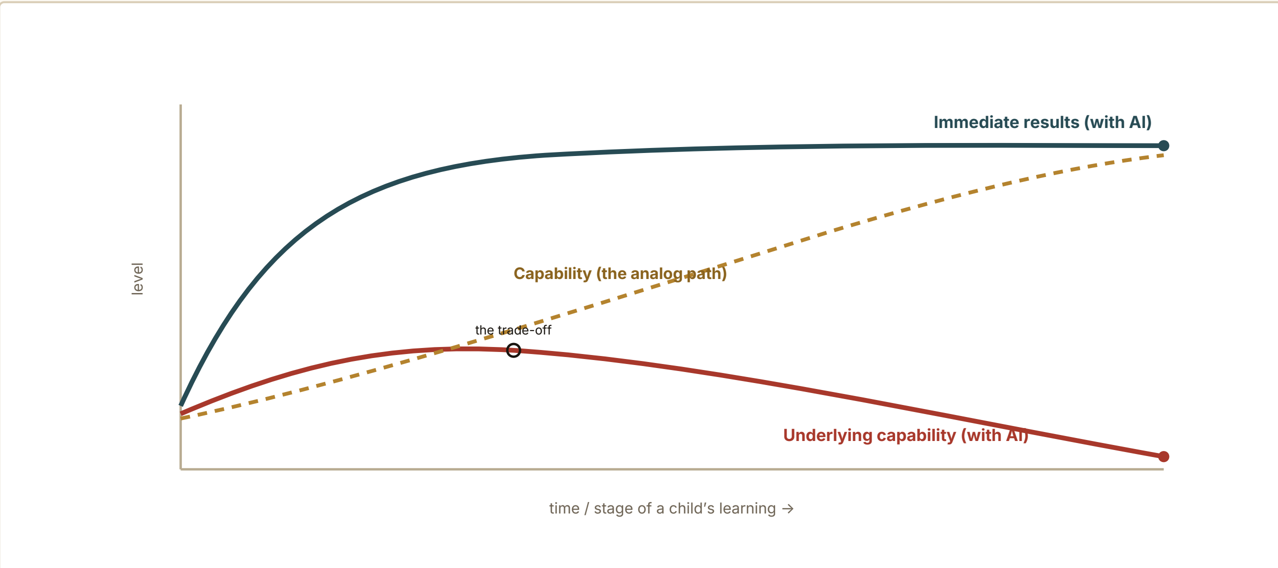

Figure 4. AI's defining trade-off. A conceptual sketch of Winkleman's claim: an **instant elevation of results** twinned with a long-term **dwindling of underlying capability** — "immediate relief in exchange for long-term dependency." The **dashed line** is the slower analog path, where capability keeps climbing. **(Conceptual, not measured data.)**

Skipping the crawling phase

Learning scientists, she notes, question AI's "deploy now, test later" posture; the industry's "move fast and break things" ethos, perhaps defensible in industry or defence, feels reckless when the thing being broken is a child's development. These systems are unpredictable. They hallucinate, presenting fabrication as fact in a way no child could be expected to unpick. They are built to keep users coming back and to tell them what they want to hear — traits we would call manipulative in a human teacher. And, she stresses, these are not teething problems awaiting a patch; they are core properties of the technology.

With some 80% of American middle schools, on her figures, now handing out Chromebooks with built-in AI that offers to fix spelling, structure and sentences, the system seems designed to skip the crawling phase — the arduous phase — of learning. But we do not tell the parents of a baby that crawling is a useless stage to be optimised away. It is the necessary prelude to walking.

§ 7 — The prescription

Tech ed, not EdTech

So what should schools actually do? Winkleman's answer, drawn from several voices at the podium, is counter-intuitive but coherent. Asked what pupils should be taught to prepare for their future, one technology futurist replied: teach the subjects that form character — classics, English, languages, history, philosophy. Skills, he argued, can be acquired later; formed minds cannot be retrofitted.

The neat slogan comes from the tech consultant Emily Cherkin: **"tech ed, not EdTech."** That is — teach how technology works, properly, in dedicated technology lessons; and return to books for everything else. Most technology skills, Winkleman points out, do not even require an internet connection, which makes them both safer and greener. And on green grounds she turns the industry's own rhetoric back on it: schools that boast of "saving paper" by ditching books then hand out electricity-hungry devices wired to water-guzzling data centres. A book — read, re-read and passed down until it falls apart — is by comparison almost evangelically green.

THE PRACTICAL CORE

Put the money into **human teachers, human examiners and physical textbooks.** Bring back handwriting, paper-based maths, and the teacher's eyes on the pupil. Confine screens to the lessons where the subject is the technology. Exempt genuine assistive needs. Everything else: analog by default.

§ 8 — The warning

Postman's two dystopias

Winkleman closes with Neil Postman's Amusing Ourselves to Death, and its famous contrast between two prophecies. George Orwell feared an oppression imposed from outside — the state that bans books and watches us. Aldous Huxley feared something subtler: that no Big Brother would be needed, because people would come to love the technologies that dissolved their capacity to think. Orwell feared those who would ban books. Huxley feared there would be no need to ban them, for there would be no one left who wanted to read one.

ORWELL · 1984	HUXLEY · BRAVE NEW WORLD
Oppression imposed from outside	No Big Brother required
Those who would **ban** books	No **need** to ban books — no one reads
We are controlled by what we **fear**	We come to **love** our oppression
The boot, the telescreen, the ministry	Autonomy surrendered, not seized

Figure 5. Two roads to the same loss. Winkleman's framing borrows Postman: the danger for our children is not the Orwellian banning of books but the Huxleyan condition in which they simply stop wanting them — seduced by devices engineered to be more immediately rewarding than thought itself.

The choice, she argues, is ours only for ourselves. If we as adults elect to surrender our own minds to our machines, that is our affair. What we must not do is inflict that surrender on children who never chose it. "Let us give them knowledge, love, truth and beauty," she ends. "They deserve nothing less."

§ 9 — Summary

In summary

Winkleman's argument moves from a playground anecdote to a civilisational warning, and its logic is cumulative. First: the persuasive mechanics we restrict on consumer apps are the same ones we wave into the classroom inside "educational" software, and they are engineered for engagement, not learning. Second: children are embodied learners, and the physical acts of reading from paper and writing by hand appear to build the brain in ways screens do not — a claim with real, if still-developing, neuroscientific support. Third: AI intensifies every one of these risks, because it can perform the very cognitive labour through which a young mind is supposed to be formed, offering immediate relief at the price of long-term capability.

The counter-arguments deserve a hearing too, and Ratty2Tails notes them plainly. Correlation between falling scores and rising devices is not proof of cause, and the same window saw social media, smartphones and a pandemic reshape childhood all at once. Some of the specific figures in the talk could not be independently verified and are presented here as its claims. Well-designed technology plainly helps some learners, and a wholesale return to chalk carries its own costs and nostalgias. Yet the direction of independent evidence — and the striking fact that country after country, and many of the tech industry's own families, are quietly walking back from screen-first schooling — gives the case more weight than easy dismissal would allow.

The prescription, at least, is refreshingly concrete: **tech ed, not EdTech.** Teach technology as a subject; teach everything else with books, paper, pens and human beings. Protect the slow, arduous, formative labour of learning to think — the "crawling phase" — from tools designed to make it disappear. Whether one accepts every figure or not, the underlying wager is hard to argue with: the capacity to think deeply may be the single human faculty an age of thinking machines can least afford to let atrophy in its children.

> "If we, as adults, choose to surrender our minds to technology, that's our choice. But we must not inflict that surrender on our children."
>
> — SOPHIE WINKLEMAN, ARC 2026

§ 10 — References & notes

References

1. **Sophie Winkleman, "Why AI in the Classroom is a Catastrophe."** Address to the Alliance for Responsible Citizenship (ARC), London ExCeL, 2026. Primary source for this essay. Video: youtube.com/watch?v=yFp5-i5fzyQ. Transcript archived at singjupost.com.
2. **F. R. (Ruud) van der Weel & Audrey L. H. van der Meer,** "Handwriting but not typewriting leads to widespread brain connectivity: a high-density EEG study with implications for the classroom." Frontiers in Psychology, vol. 14 (2024). DOI: 10.3389/fpsyg.2023.1219945. — Underpins the handwriting-versus-typing claim in §3.
3. **Sweden's textbook reinvestment.** Multiple reports (2024–2026) document the Swedish government committing on the order of €100 million+ to reintroduce printed textbooks after PISA declines, with Denmark, Finland and Norway moving in the same direction. See, e.g., UNESCO Courier, "Sweden: the unmet promises of the digital classroom" (2026); Undark, "Why Swedish Schools Are Bringing Back Books" (2026).
4. **Neil Postman,** Amusing Ourselves to Death: Public Discourse in the Age of Show Business (Viking, 1985). Source of the Orwell-versus-Huxley framing in §8.
5. **Jonathan Haidt,** The Anxious Generation (2024), and related public commentary on removing devices from classrooms — referenced in the talk.
6. **Richard H. Thaler & Cass R. Sunstein,** Nudge (2008), on the human default to low-effort choices — cited in §6.
7. **OECD PISA analyses** linking intensive in-class device use to weaker outcomes (2015; 2023–2024 reporting). Context for the paper-versus-screen claim in §4.

A note on names and figures. Several individuals and one experiment quoted in the address appear only in the spoken transcript and, in some cases, with names that seem to have been garbled in transcription. Where Ratty2Tails could not confirm a source, the relevant claim is attributed to the talk itself and flagged accordingly. Statistics such as "one million UK children" and "80% of US middle schools" are reproduced as stated by the speaker.

Colophon. Edited and annotated by **Ratty2Tails** as part of the ongoing essay series. Set in **Fraunces** (display), **Source Serif 4** (text) and **Inter** (labels). Diagrams hand-authored as inline SVG; typeset with WeasyPrint. British English throughout. This edition faithfully adapts a public address for reading; interpretation, emphasis and editorial hedging are the editor's own, and any errors introduced in adaptation are likewise the editor's.